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1864 Prep — User Guide

Get any agency's data ready to join the network

ATTACHMENT 1 · USER GUIDE

1864 Prep turns any messy government or survey file into clean, trustworthy, shareable data — on your own computer, with every change shown for your approval. This guide explains what it does, why it matters, and exactly how to try it — no installation required.

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1. Why standardisation comes first

Before an agency can answer questions on a shared network, its data has to be legible — consistent dates, intact identity numbers, recognisable place names, no scrambled text. In practice, most agencies keep their records in spreadsheets that have grown organically over years: merged headers, notes above the table, mixed date formats, duplicate rows, and long ID numbers that lose their leading zeros the moment Excel touches them. None of that is a failing of the staff; it is simply what real administrative data looks like.

1864 Prep exists to close that gap without asking anyone to change how they work. It reads whatever an agency actually has, works out what each column means, proposes fixes a person approves one at a time, and produces a tidy file plus a complete record of every change. It is the groundwork that lets verified answers travel across an open network — while the raw records never leave the user's machine.

Try it now, nothing to install: open one864prep-demo.onrender.com and run a messy file through it. Source code is open at github.com/abiola1864/1864Prep.

2. The two promises that never change

Promise 1 — Your data stays with you

The engine runs locally; nothing is uploaded. The hosted demo uses synthetic data only, and the optional AI assist is off by default. When you run 1864 Prep on your own machine, your file never leaves it.

Promise 2 — Meaning is never changed without your sign-off

Safe formatting (spacing, capitalisation) is applied and listed for you; anything that could change meaning is a proposal you accept or reject. Every decision is recorded in a change log you can keep and share.

3. What 1864 Prep does

1864 Prep first works out the **structure** of a file before touching a single value: it finds the real table (skipping titles, logos, and blank rows), resolves multi-row and merged headers into one clear name per column, picks the right sheet in a multi-sheet workbook, and reads the whole column before deciding its type — so one odd cell cannot mislead it. Only then does it look at the values.

THE INTELLIGENT PART — READING MESSY DATA LIKE AN ANALYST

Capability	What it does

Interprets what each field means	Works out on its own whether a column is a date, a name, an amount, or a code.
Finds information in free text	Detects people, places, organisations, and money inside free-text fields.
Reads most files and broken headers	CSV, Excel (.xlsx / .xls), JSON, and PDF tables — any encoding, even multi-row and merged headers.
Fixes mismatched but identical labels	Links categories spelled differently that mean the same thing.
Matches values to official registers	Aligns entries to the correct official name (states, LGAs, countries, currencies) and suggests the fix.

THE DEPENDABLE PART — GROUNDWORK THAT MAKES DATA TRUSTWORTHY

Capability	What it does
Standardises dates and numbers	One consistent format for dates, numbers, phones, and emails across the file.
Repairs encoding and date errors	Recovers scrambled text and Excel date-serials (44562 → 2022-01-01) that other tools miss.
Preserves ID and account numbers	Keeps leading zeros and long IDs intact — the key to matching records across agencies.
Flags duplicates and outliers	Catches repeated rows and impossible values, and asks before acting.
Logs every change for approval	Nothing is altered without a person’s sign-off, and all of it is recorded.

4. How to try it — a first run in five steps

The whole flow, from a messy upload to a clean export with a change log:

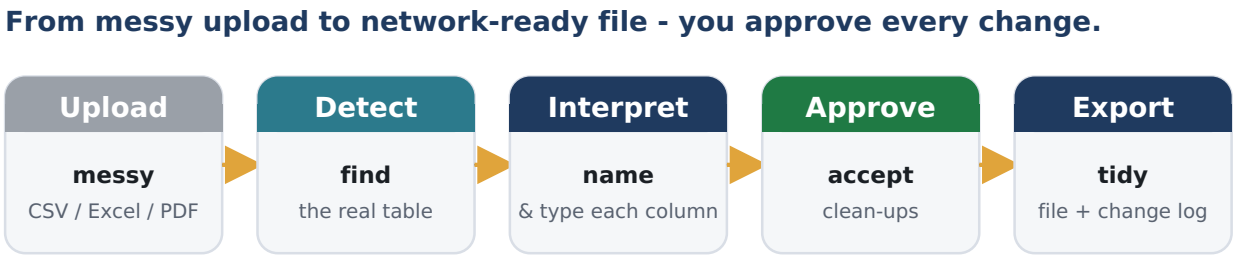


Figure: the 1864 Prep clean, end to end.

1. **Open the demo.** Go to one864prep-demo.onrender.com. No account or install is needed.
2. **Upload a file.** Drag in a messy spreadsheet, or use a provided sample. 1864 Prep detects the real table and reports what sat above the header.
3. **Review the reading.** It proposes a clean name and a type for every column. Correct anything it gets wrong — you are always in control.
4. **Approve the clean-ups.** Each proposed change appears as a worklist item you accept or reject. Safe formatting is applied and listed; meaning-changing edits wait for you.
5. **Export.** Download a tidy CSV, Excel, or Word file together with a complete change log recording every decision.

See your data before and after. For any numeric column, 1864 Prep can show the shape of your data — an interactive histogram with mean, median, and flagged outliers — on both the raw and cleaned data, side by side, so you can see exactly what cleaning recovered.

5. The wider toolkit

Beyond the guided clean, 1864 Prep includes focused tools, each with its own step-by-step workflow:

Tool	What it is for
Find outliers	See the spread, choose a method, review flagged values, decide keep / cap / remove.
Find & remove duplicates	Pick match columns, check confusing ones, review each group, choose which to keep.
Validate data	Set required fields and rules, review issues, export a report.
Match & merge files	Confirm a shared key, choose a join type, preview, export.
Anonymise / mask	Hash, mask, or drop sensitive columns, with a before/after preview.
Summarise / profile	Quick statistics and a readable profile of any dataset.
Compare two files	Confirm a row key, review differences between an old and a new version.
Combine / append	Check columns align across files, preview, export.

6. The optional AI assist — private by design

The base tool is fully offline. If an API key is present *and* the user switches it on, an optional AI assist can help name or categorise a tricky column. It is built so the model never sees the dataset:

- It works **one column at a time**, sending only a few **distinct** sample values — never whole rows, never row counts.
- Sensitive columns (phone, ID, name, account) are **masked to their shape** (08031234567 → 99999999999) or reduced to a pattern, or nothing is sent at all.
- The interface shows a clear **online/offline switch** and can **preview exactly what would be sent** before anything leaves the device.

> **Default posture:** off. 1864 Prep is fully functional with the AI assist disabled; it is an optional convenience, never a requirement, and it is designed to be safe to turn on even for sensitive registers.

Under the hood, for the technically minded

The engine is organised into clear stages: ingestion (file reading, sheet selection, header resolution, encoding detection); profiling (per-column type inference); a set of consent-first value transforms; official reference data for matching; safe near-duplicate detection; a distribution engine for the before/after view; and exporters for CSV, Excel, and Word with the change log. Around twenty automated test suites cover these components. Installation for local use is a single command:

```
pip install -e .            # core engine
uvicorn app.server:app --reload   # then open http://localhost:8000
```

7. How it connects to the Open Network

1864 Prep is the first half of a two-part design. It gets an agency's data ready; the Open Network for Data Exchange then lets that agency answer verified questions from other agencies — without the data ever leaving its machine. An agency cleans its file once, locally, and is then ready to participate.

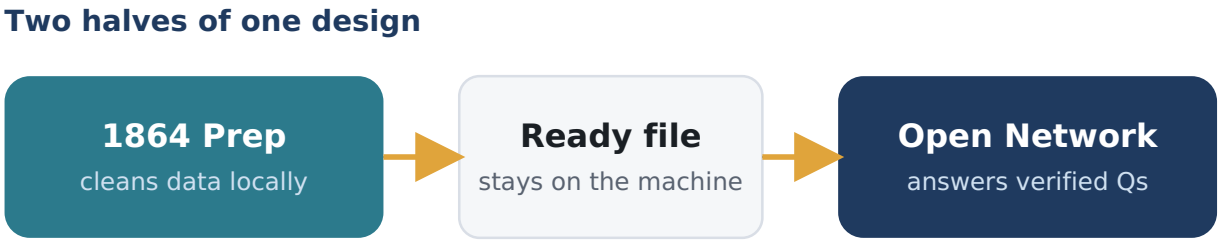


Figure: clean once, locally; then answer on the network - data never leaves.

The companion **Open Network User Guide** explains how an agency connects and answers, and the **Technical Documentation** covers the architecture in full.

Built by the 1864 Research and Innovation Institute. Open source — contributions and issues welcome at github.com/abiola1864/1864Prep.